The Contaminated City

Chief Scientific Officer · Riverton Contamination Response

15 DAYS · 52 STOPS · GRADE 12 · CONTAMCITY

THE OPENING CARD

A freight yard burned last night, 900 metres up the river from Riverton's water intake. The intake is shut and the whole city is living on stored water. You are the Chief Scientific Officer, so no valve opens until you say it can. In fifteen days you sign The Riverton Evidence Package: what escaped, where it went, and what must keep working. Get it wrong and the city drinks what burned.

DAY 1 OF 15

What Burned?

BEFORE THE DAY — GREET

Six leads, one incident, and nobody has worked together before

The response was assembled on Tuesday out of whoever was reachable, and by Friday somebody will be signing a release based on a number a stranger produced. Okonjo, the analytical chemistry lead, wants you known to as many of the leads as you can before the first sample comes in, because a result you cannot attach to a person is a result you will quietly re-run.

PLAN CARD 47W

On day 1 the fire is out, but seven damaged drums sit 80 metres from the river. The radio is already passing three different names for what is inside them. Today you say what is known, what is still a guess, and which test to buy first.

BRIEFING 20W

Fire Command needs to know which materials can meet water, which need isolation, and which identities are still only guesses.

WHAT YOU DECIDE 14W

Turn damaged cargo, records and first laboratory measurements into a defensible provisional hazard list.

1.1 Read the formula, not the rumor

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · PROTOCOL

SCENE 42W

A freight-yard fire has damaged several unlabelled containers, 80 metres from the river. Okonjo has written all three radio guesses on the whiteboard and crossed out none of them yet. Firefighters are picking a suppression agent from whatever identity you give them.

GUIDE 60W

The left column is four kinds of record. The right column is four kinds of claim. Ask of each record what it actually fixes: which elements, in what ratio, carrying what charge. A trade name fixes none of the three. Some substances on the shortlist react with water. A firefighter is choosing a suppression agent from whatever identity comes back.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a chemical formula lists which elements are present and in what ratio · atomic structure, ions and electron configuration — taken as read · periodic trends and what an element can do — taken as read

VERDICT 86W

A formula is not a name. It states which elements are present, in what whole-number ratio, and what charge the unit carries — and each of those constrains what the substance can do. Change a subscript or a charge and you have a different compound with different reactivity, different solubility and a different hazard class. A common name carries none of that, which is why a label reading only a trade name cannot support a hazard prediction. Some of the wrong answers here react with water.

TAKEAWAY 8W

Chemical identity begins with explicit composition and charge.

1.2 From damaged container to provisional identity

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · SEQUENCE

SCENE 44W

Inside the mobile laboratory, Okonjo sets the first intact sample from the damaged drums on the bench, still sealed. The outside markings are legible now and will scuff off on opening. Drawing liquid vents the headspace, and there is only enough for two runs.

GUIDE 69W

Every one of these four is available right now, so nothing here is a clock. What separates them is what you can still do afterwards. Ask of each step what it consumes: the markings, the headspace, the liquid, or the sample itself. There is enough liquid for two runs. A step that ends the sample answers best and answers once, which puts it somewhere very particular in the order.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

some measurements leave the sample as they found it and some use it up

VERDICT 85W

The order is set by what each step costs. The markings on the outside of the drum survive only until somebody opens it, so they are recorded first and consume nothing. Headspace vapour samples what has already escaped, without breaching the bulk liquid. A non-destructive spectrum on a small aliquot can be repeated all week if it goes wrong. The destructive method gives the best identification and gives it once. It is therefore the last thing you spend and the first thing you would regret.

TAKEAWAY 12W

Analytical work runs from what costs nothing to what cannot be undone.

1.3 Spend the first of the lab budget

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · VALUE

SCENE 42W

An $800 laboratory budget is left for the first round of evidence. Three damaged drums have lost their labels, and Okonjo confirms the first extract is ready. Firefighters wait at the fence line for the provisional identity that decides their next move.

GUIDE 62W

Open each evidence card to see what it would actually tell you, then spend up to $800. A card's price is not its value. Firefighters at the fence need an identity firm enough to choose their next move, so buy the evidence that could distinguish between the candidates. Repeating a test you have already run buys certainty about something you already knew.

ASK 11W

Which evidence do you buy before giving firefighters a chemical identity?

BEHIND THE BUTTON 136W

What makes a test worth its money. Not precision, and not how much data it returns. The question is whether its two possible results point at different chemicals. A test every candidate would pass tells you nothing you did not know before you paid for it.

Why a provisional identity is enough. The firefighters do not need a publication-grade answer; they need to know which class of chemistry they are standing next to. Waiting for certainty means they act on nothing at all, which is the more dangerous of the two errors.

Why the labels being gone matters so much. Three drums, no labels, and one extract means the identity has to come out of the chemistry rather than the paperwork. Every card on the board is a way of asking the material what it is.

TAKES AS READ

different analytical methods can fail for different reasons · a repeated measurement is not automatically an independent measurement

VERDICT 78W

An analytical result can be precise and still be wrong. A repeat run tests repeatability, records test provenance, and quality controls test specific laboratory failure modes. None of those independently identifies the compound. A structural method based on different physics can overturn a mistaken retention-time match, so it changes the decision directly. With a limited budget, the first purchase should attack the uncertainty that can reverse the hazard call. Making the existing result look more complete is secondary.

TAKEAWAY 20W

Independent evidence is more valuable than a repeated result when the decision depends on whether the identity itself is right.

1.4 Water first, or never water

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · BELT

SCENE 33W

Fire Command has hose crews ready, but the mixed cargo contains both ordinary soluble materials and substances for which water can create heat or flammable/corrosive products. The drums cannot be sorted by appearance.

GUIDE 46W

Two bins. Something that simply dissolves is diluted by water and made less dangerous by it. Something that reacts with water makes heat, gas or both, so water is what turns a spill into an event. Sort on what the substance does when water reaches it.

ASK 14W

Send each drum to the bin that says what water would do to it.

BEHIND THE BUTTON 91W

Why this is the first question at a yard. The default response to almost anything is water — hoses are what a fire crew has. For most of what is here that is right, and for a short list it is the one action that makes the situation worse.

What reacting with water looks like. Alkali metals and hydrides give off hydrogen. Concentrated acids give off enough heat to boil what is poured onto them. Carbides and some chlorides give off gas that is worse than what was in the drum.

TAKES AS READ

some substances dissolve in water and some react with it · atomic structure, ions and electron configuration — taken as read

VERDICT 78W

The question is not whether a substance merely dissolves; it is whether contact with water creates a new acute hazard. Many salts and dilute aqueous solutions can be diluted safely under controlled response conditions. Alkali metals, hydrides, carbides and several reactive chlorides generate heat or flammable/corrosive gases. Concentrated sulphuric acid and quicklime can release dangerous heat on wetting. Powdered magnesium can also generate hydrogen. The labels matter because the same hose that helps one spill can worsen another.

TAKEAWAY 14W

The default response is right for most things and catastrophic for a short list.

END OF THE DAY 11W

Create a provisional identity list and choose the next discriminating measurements.

DAY 2 OF 15

The Invisible Cloud

PLAN CARD 48W

On day 2 the wind turned forty degrees at dawn, and the vapour now drifts toward two streets of homes. The cloud cannot be seen, and police hold the exit roads open. Today you use the gas-law volume for scale, then pick the readings that move the line.

BRIEFING 16W

Gas laws set the scale of the volatile release; wind, terrain and chemistry decide the exposure.

WHAT YOU DECIDE 18W

Bound the volatile release and set evacuation or shelter triggers from measurements rather than from gas-law volume alone.

2.1 How much volume can the gas occupy?

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · BALLPARK

SCENE 43W

At the command trailer, a colourless plume is moving toward two neighbourhoods. Police are holding the evacuation roads open while the wind shifts. The release inventory is finally known, and the incident commander, Grace Boateng, wants the gas scale before moving her roadblocks.

GUIDE 62W

Six numbers, and two of them describe conditions the release is not at. One is the molar volume at standard temperature and pressure. One is zero Celsius in kelvin. The relationship is written for ambient conditions, so every term has to be the ambient one. A volume computed at the wrong conditions still looks like a volume. Roadblocks get moved on it.

ASK 13W

Use the ideal gas law to estimate the gas volume at ambient conditions.

BEHIND THE BUTTON 134W

What the relationship says. $V = nRT/P$, with V the volume, n the moles of gas, R the gas constant, T the absolute temperature and P the pressure.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a given amount of gas occupies a predictable volume at ordinary conditions

VERDICT 88W

The ideal gas law turns an amount of gas into a volume you can reason about. At ordinary temperature and pressure, the scale is about 24 litres per mole. It is a first-order tool and nothing more: it says how big the release is, not where it goes. The hazardous footprint depends on mixing, wind, terrain and chemistry, none of which appear in the calculation. The number still has to be right, because evacuating too little exposes people and evacuating too much blocks the roads the response needs.

TAKEAWAY 10W

Simple gas laws provide scale while atmospheric transport determines exposure.

2.2 Interpret plume behavior

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · PROTOCOL

SCENE 44W

Varga leads atmospheric chemistry. At the weather station he reads out rising temperature, falling pressure and a strengthening afternoon wind. The plume is still moving. Ground crews also report vapour pooling in a low drainage channel on the downwind side of the freight yard.

GUIDE 69W

Two different physics run at once here, and the four situations do not all belong to one. Some change what a fixed amount of gas occupies. Others change where the parcel goes. Ask which of the two each one acts on before pairing it. Read a transport change as a gas-law change and the corridor is drawn from the wrong physics. The vapour is pooling in a drainage channel.

ASK 32W

Kinetic molecular theory sets what the gas itself does; the wind sets where it goes. Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a gas has a state and also a place, and they change for different reasons

VERDICT 62W

Gas-law behaviour and plume transport answer different questions. Higher absolute temperature increases the equilibrium volume of a fixed amount of gas at fixed pressure; lower pressure does the same. A vapour denser than ambient air may initially collect in low spots, but mixing and terrain can rapidly dominate. And a wind shift repoints the downwind corridor whether the chemistry changed or not.

TAKEAWAY 9W

A plume is a coupled thermodynamic and transport problem.

2.3 Why the drum is worse in the afternoon

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · CHOICE

SCENE 44W

The same open drum reads three times the vapour concentration at two in the afternoon as it did at six in the morning. Nothing has been added and nothing taken out. Varga logged both readings himself and wants them explained before the next briefing.

GUIDE 72W

All four options would explain a rising reading, and they differ in what has changed. One changes the liquid, one changes the instrument, one changes the amount, and one changes only the partition between liquid and air. Nothing was added and nothing was removed, which is the fact that does the work. Getting this wrong sets an exposure limit from a dawn reading and clears a site that is worse by mid-afternoon.

ASK 9W

Same drum, same liquid, three times the vapour. Why?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a liquid and its vapour sit in equilibrium at a temperature

VERDICT 88W

Vapour pressure is the pressure of gas in equilibrium with its own liquid, and it climbs steeply with temperature — often doubling for every 10 or 15 degrees. Warming the drum from morning to afternoon moves more molecules into the gas phase above the same liquid, so the air above it is more concentrated with no change in what is in the drum. This is why exposure limits are checked at the hottest part of the day, and why a hazard assessment made at dawn understates the afternoon.

> **TAKEAWAY** 18W
>
> How much of a liquid is in the air above it depends steeply on how warm it is.

END OF THE DAY 11W

Provide a bounded plume assessment and decision triggers for evacuation zones.

Do Not Open the Hatch

PLAN CARD 49W

On day 3 two dirty drains vanish into the same closed tunnel, and a repair crew waits at the hatch. Nobody sends them down on chemistry that has not been balanced. Today you bound the most gas and heat that reaction could make, then ask whether it is running.

BRIEFING 25W

Runoff from two parts of the mixed cargo may be meeting in a closed drainage tunnel, and the chemistry must be bounded before anyone enters.

WHAT YOU DECIDE 15W

Bound the largest plausible underground reaction and decide what evidence is required before confined-space entry.

3.1 From identity to reaction estimate

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · SEQUENCE

SCENE 45W

At the tunnel entrance, two drainage streams may be meeting behind a closed access door. Oyelaran, the confined-space safety officer, holds his repair crew at the barricade, and the storm bypass remains shut. The laboratory has analyses from both drains but no reaction estimate yet.

GUIDE 66W

These four are not four independent tasks. Each needs the output of another. Ask of each card what it requires before it can be done at all. Coefficients cannot be used until the equation is balanced. Masses cannot be compared until they are moles. A prediction is worth making only once there is a measurement to compare it against. A repair crew is at the barricade.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a balanced equation says how many of one thing react with how many of another

VERDICT 74W

Every quantitative reaction prediction begins with a balanced equation. Its coefficients supply the mole ratios between reactants and products. Measured masses are then converted to moles, because the coefficients compare particles rather than grams. Those mole ratios identify which reactant runs out first. Only after the limiting reactant is known can the maximum products be calculated. The prediction can then be checked against the gas, heat or remaining reagent actually observed in the tunnel.

TAKEAWAY 10W

Stoichiometry is a conditional prediction based on a specified reaction.

3.2 Which reactant limits?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · BALLPARK

SCENE 39W

A reaction takes two moles of A for every mole of B, and both are arriving in the tunnel in amounts nobody chose. Tomas Brandt, who leads reactions and energy, will not clear an entry until this number exists.

GUIDE 65W

Four numbers, and one of them is the two reactant amounts added together. Another is the reaction's own ratio rather than a quantity of anything. A balanced equation compares particles, so amounts have to be in moles before the ratio can be applied. Adding two reactants gives a number that no stoichiometry uses, and closed-space entry decisions are being made from whichever number comes out.

ASK 14W

Which reactant limits, how much B can react, and what percent yield was recovered?

BEHIND THE BUTTON 133W

What the relationship says. First use $n = m/M$. Then compare mole amounts with the 2 A : 1 B coefficients. Percent yield = actual/theoretical × 100.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a reaction stops when one ingredient runs out

VERDICT 72W

A balanced equation compares particles, so masses have to become moles before the reactants can be compared. 500 grams of A at 50 g/mol is 10 mol. 640 grams of B at 80 g/mol is 8 mol. The 2:1 ratio means 10 mol A can consume only 5 mol B, so A limits. If the recovered product corresponds to 4.2 mol of reaction, the percent yield is $4.2/5.0 \times 100 = 84\%$.

TAKEAWAY 17W

Stoichiometric limits are set in moles, and percent yield compares what was recovered with that theoretical limit.

3.3 Heat release is not the spontaneity test

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · BALLPARK

SCENE 43W

A clean-up reagent is staged beside the tunnel; the tank test shows no product after 6 days at 18 °C. Léa Moreau, the energy division's researcher, hands over the measured ΔH and ΔS and asks whether the chemistry is impossible or simply slow.

GUIDE 69W

Four numbers, and two of them are the same temperature written in different scales. The relationship needs an absolute temperature, so degrees Celsius put straight into it is not a small error but a different quantity. Note what the answer does and does not settle: a favourable sign says the reaction can go, not that it will go this week. The tank has had six days and shows nothing.

ASK 24W

At 18 °C, what is ΔG — the free energy change — and what does the lack of product say about the reaction rate?

BEHIND THE BUTTON 119W

What the relationship says. $\Delta G = \Delta H - T\Delta S$, with T in kelvin.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

temperature in $\Delta G = \Delta H - T\Delta S$ must be in kelvin · a negative ΔG means the forward reaction is thermodynamically favored

VERDICT 70W

Free energy combines enthalpy, entropy and temperature. At 291 K, $\Delta G = -180 - 291(-0.40) \approx -64$ kJ/mol, so the reaction is thermodynamically favorable. That does not make it fast. The unchanged tank points to a substantial activation barrier at this temperature. This distinction matters in treatment design. Temperature or a catalyst can change reaction rate without changing the sign of ΔG. A favorable ΔG alone never guarantees useful speed.

TAKEAWAY 14W

Free energy sets thermodynamic favorability; activation barriers set how quickly a favorable reaction proceeds.

3.4 Read the formula, not the rumor — Review

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CALLBACK · PROTOCOL

SCENE 39W

Hiroshi Tanaka, the records and shipping clerk, has pulled four entries for a second consignment that stood in the same shed. Each line describes its drum differently. Okonjo wants them sorted before the bench spends any sample on them.

GUIDE 67W

Four entries, and they do not constrain the same amount. Ask of each what it actually fixes: which elements, in what ratio, carrying what charge, or none of the three. Two of them fix a ratio and disagree about which ratio they have handed you. A drum sent to the bench on a line that fixes nothing costs a run of sample nobody can go back for.

ASK 17W

Match each paperwork entry to what it actually fixes about the drum. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

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TAKES AS READ

a chemical formula lists which elements are present and in what ratio · atomic structure, ions and electron configuration — taken as read

VERDICT 82W

A formula is a claim about composition, and each of these entries makes a different amount of that claim. A name without the metal's charge leaves two salts open, and they differ in solubility and in what they react with. Percentages by mass fix the ratio of atoms and therefore the simplest formula, but not how many of that unit a

> molecule holds. Subscripts sharing a divisor say the opposite. A transport number sorts a drum for carriage and fixes nothing chemical.
>
> **TAKEAWAY** 12W
>
> A record constrains a substance only as far as it states composition.

END OF THE DAY 16W

Estimate the maximum reaction scale and define measurements that verify whether the assumed chemistry is occurring.

One Result Is Lying

BEFORE THE DAY — FOLLOW

Ferreira walks the river reach once

Where the plume actually enters the river is not on any drawing at the accuracy anybody needs, and Ferreira, the water and sediment lead, can read it off the bank — the colour of the silt, where the weed stops. She walks it once this morning. Stay with her, close enough to see what she is pointing at and far enough back not to be standing in it.

PLAN CARD 49W

On day 4 the lab has enough data to name the main parts of the cargo. Two runs of one method agree well, and a second method does not agree at all. Today you say which result is real, and whether two matching numbers are one error made twice.

BRIEFING 18W

The chromatogram contains more peaks than the manifest explains, and two apparently agreeing measurements share a hidden dependency.

WHAT YOU DECIDE 19W

Confirm the major cargo components and identify a common-mode laboratory failure before the wrong compound enters the treatment plan.

4.1 Which alkali metal is the greater water hazard?

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CHOICE

SCENE 27W

Records narrow two unlabelled metal cans to sodium and potassium. Fire Command needs to know which one demands the larger water-exclusion zone before the cans are moved.

GUIDE 37W

The records have already narrowed the problem to two elemental alkali metals. Ask what changes down group 1: shielding rises and ionisation energy falls. That trend is useful here precisely because the chemical form is held fixed.

ASK 21W

Two recovered metal samples are sodium and potassium. Use periodic trends to rank their water reactivity before either container is opened.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

both samples are elemental alkali metals under comparable conditions · atomic structure, ions and electron configuration — taken as read

VERDICT 58W

For elemental alkali metals, the outer electron is farther from the nucleus and more shielded going down group 1, so ionisation energy falls. Potassium therefore reacts more vigorously with water than sodium under comparable conditions. The trend applies to the elemental metals; it is not a licence to rank every compound containing sodium or potassium the same way.

TAKEAWAY 23W

> A periodic trend predicts behaviour for elements held in the same chemical form, and it stops predicting anything the moment the form changes.

4.2 Read analytical disagreement

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · TRACE

SCENE 46W

Two methods have been run on the same extract and they do not tell the same story. Yusra Haddad, the independent reviewer, flags the disagreement — a false identification here sends the plant a chemistry that could create a worse product than the one it removes.

GUIDE 50W

Open each channel to see what its result depended on. Keep the ones that stand on an independent measurement, and untick the ones whose chain runs through a shared step. Then name the step they have in common. Two methods disagreeing is information, not an error to be averaged away.

ASK 11W

Which channels remain trustworthy, and what shared source is at fault?

BEHIND THE BUTTON 143W

Why disagreement is the useful signal. Two methods agreeing can mean both are right or both inherit the same mistake. Two disagreeing means at least one assumption is wrong, and finding which is a question about what each method depended on rather than about which number looks better.

What a shared step looks like here. A calibration standard, a dilution, a column, an internal reference: anything both methods passed through. Everything downstream of it moves together if it is wrong, which is why the average of two such results is no safer than either.

What is riding on getting it right. A false identification sends the plant a treatment chemistry chosen for the wrong compound, and the product of that reaction can be worse than what it removes. Withdrawing both results is safer than averaging them and worse than naming the shared step.

TAKES AS READ

a blank contains every preparation step except the sample · independent methods do not share the same critical failure path

VERDICT 80W

The sample and duplicate chromatograms agree because they share the same preparation solvent, retention reference and detector path. Their agreement therefore does not provide two independent identifications. The blank carries almost the same 3.1-minute feature even though it contains no river sample, which points to preparation contamination. The structural spectrum uses a different measurement path and does not match the target. Together, the blank and the independent spectrum overturn the retention-time identification while leaving the rest of the run interpretable.

TAKEAWAY 16W

> Agreement counts as independent evidence only when the measurements do not inherit the same failure path.

4.3 Resolve the ambiguous peak

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A PERSON · CHOICE

SCENE 43W

1 peak at 4.2 minutes matches 2 candidates to within 0.05 minutes, and Ana Kowalski, who keeps the reference library, has both on the shortlist. The river sample also carries humic matter, chloride at 40 mg/L, and a detergent from the fire foam.

GUIDE 62W

Two candidates come off the column together. So anything that depends on how fast they travel inherits the ambiguity. Sort the options by that test. Does this one improve the separation, replace it with evidence of another kind, add controls, or only change the presentation? Controls make a number defensible and cannot resolve an overlap. Better presentation adds no measurement at all.

ASK 23W

One peak is still ambiguous, in a river sample carrying chloride at 40 mg/L and detergent from the fire foam. What settles it?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

two compounds that overlap in one method may not overlap in another

VERDICT 84W

There are two honest ways out of an overlap. Improve the separation until the components stop coming off the column together, or bring in evidence of a different kind so the answer no longer depends on the separation at all. The second is faster, and it cannot inherit the problem: a method that identifies by structure does not care that two compounds travelled at the same speed. The third way is to present the same weak data more attractively, and it adds no measurement.

TAKEAWAY 15W

Better presentation cannot repair a method that was fooled, or evidence that was never validated.

4.4 How much volume can the gas occupy? — Review

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · CALLBACK · BALLPARK

SCENE 41W

Erik Lindqvist, the dispersion modeller, has a second cylinder logged in the vented store on the north side of the yard. Simran Kaur, who coordinates evacuation, will not lift the shelter order until somebody says how much air that inventory becomes.

GUIDE 66W

Six numbers, and two of them belong to conditions the store is not at. One is the Celsius figure on the thermometer, which is a reading rather than an absolute temperature. One is a molar volume quoted somewhere else. Every term in the relationship has to be the one measured in that store. A shelter order over two streets is being held on whatever comes back.

ASK 17W

Use the ideal gas law to estimate the gas volume at the conditions measured in the store.

BEHIND THE BUTTON 134W

What the relationship says. $V = nRT/P$, with V the volume, n the moles of gas, R the gas constant, T the absolute temperature and P the pressure.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a given amount of gas occupies a predictable volume at stated conditions

VERDICT 86W

The ideal gas law converts an inventory into the volume it would fill at the conditions it is at, and that is all it does. The temperature and the pressure in the store are both off the round figures. The relationship takes the measured ones, or it answers a different question. Temperature has to be absolute, so degrees Celsius put straight in is not a small error but a different quantity. What comes out is a scale for the release rather than a footprint for it.

TAKEAWAY 21W

A gas-law volume is a scale for the release, and it is only as good as the conditions put into it.

END OF THE DAY 9W

Assign confidence-ranked identities to mixture components using independent evidence.

Where Did the Metal Go?

PLAN CARD 46W

On day 6 the water at the intake looks better, but the metal has not left the river. Some is dissolved, some is stuck to particles, and some is in the bed. Today you choose where to sample, so cleaner water cannot be read as clean.

BRIEFING 14W

A clean surface-water grab cannot show whether the dissolved-metal fraction disappeared or merely moved.

WHAT YOU DECIDE 16W

Predict how the dissolved-metal fraction partitions among water, particles and sediment, then design a mass-balance survey.

5.1 Where will the chemical go?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · PROTOCOL

SCENE 43W

Runoff from the accident has reached the river, and Ferreira is on the bank with empty sample bottles, waiting on a prediction. Choose the wrong phase and she samples the wrong thing — a clean water result the city reads as an all-clear.

GUIDE 69W

The left column is four molecular structures and the right column is four places a chemical can end up. Pair them by asking what water can do for that structure: hydrate an ion, hydrogen-bond a polar molecule, or refuse a non-polar one. A compound that avoids water sits on the particles instead. Sample the wrong phase and a clean water result gets read by the city as an all-clear.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

like dissolves like — polar mixes with polar, and non-polar does not · atomic structure, ions and electron configuration — taken as read

VERDICT 81W

A chemical released into water does not stay in one place or one phase, and where it goes follows from molecular structure. Polar molecules that can hydrogen-bond dissolve readily. Ionic species dissolve as hydrated ions, surrounded by water molecules. Non-polar liquids, which barely mix with water, separate out or spread as a surface film. Compounds that avoid water partition onto organic carbon in suspended particles and sediment and stay there. Absence from the water column is not absence from the river.

TAKEAWAY 13W

The absence of a chemical from water does not mean it has disappeared.

5.2 Use the common-ion effect to clear the limit

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · CHOICE

SCENE 44W

The filtered water has 3.1 mg/L of dissolved metal against a 0.5 mg/L permit limit. The solid is a 1:1 metal carbonate and pH is fixed. Osei, the treatment engineering lead, can raise dissolved carbonate eightfold at the pilot without changing the water volume.

GUIDE 65W

All four options give a number and a reason. The reasons are what differ: fixed solubility, a product held constant, both ions rising, or a squared dependence. Check the stoichiometry of the solid first. The exponents in the solubility product come from it. This has to clear a permit limit. An answer out by a factor of eight is a discharge that fails its consent.

ASK 10W

Eight times the carbonate: what happens to the dissolved metal?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a saturated solution is in equilibrium with a 1:1 solid · Ksp stays constant when temperature stays fixed

VERDICT 72W

For a 1:1 metal carbonate, the solubility product is proportional to the dissolved metal concentration times the carbonate concentration. At fixed temperature, that product stays constant at equilibrium. Raising carbonate by a factor of eight therefore drives the dissolved metal down by the same factor. Starting from 3.1

mg/L gives about 0.39 mg/L. That is below the 0.5 mg/L limit, although a real pilot still has to check pH, complexation and settling.

TAKEAWAY 19W

For a 1:1 sparingly soluble salt, increasing 1 ion lowers the equilibrium concentration of the other in inverse proportion.

5.3 Spend the river survey budget

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · ALLOCATE

SCENE 38W

Ferreira has 20 samples in the survey budget. The compound is sparingly soluble, and the river runs 900 metres from the release to the intake. Her two grab samples taken so far disagree by a factor of six.

GUIDE 58W

Twenty bottles is the whole survey. Each package you select spends some of them, and the panel shows which river questions your current plan can answer. Watch that list rather than the map. Two grab samples already disagree by a factor of six, so a plan that cannot explain why is a plan that will be argued with.

ASK 13W

How do you spend 20 bottles without leaving a required river question unanswered?

BEHIND THE BUTTON 148W

Why the disagreement is the design brief. A factor of six between two grabs means the compound is not evenly mixed. That can be plume structure across the channel, depth stratification, or a change with time. Each of those is a different sampling pattern, and one grab per site cannot tell them apart.

Why sparing solubility matters. A compound that barely dissolves travels partly as droplets or on particles rather than in solution, so a filtered bottle and an unfiltered one answer different questions. Nine hundred metres is far enough for that difference to matter at the intake.

What the required questions are for. Not everything the survey could ask, but the ones a decision hangs on: whether the intake is affected now, whether it will be, and by how much. A plan that answers other things beautifully and misses one of those has spent twenty bottles badly.

TAKES AS READ

different sample locations and phases answer different questions · a clean water sample cannot rule out contaminant stored in solids

VERDICT 83W

Environmental sampling needs coverage across the dimensions that can change the conclusion. Intake samples measure current exposure, upstream samples establish background, and solids plus sediment test whether a water-avoiding compound has left the water column. A visually stained bank may be interesting, but it is not required for the main decision. The survey must test movement toward the intake and storage in another phase. The twenty-bottle limit forces those priorities to be explicit instead of letting every desirable sample appear on the plan.

TAKEAWAY 22W

A sampling plan is a set of questions it can answer, and every bottle spent on one question is unavailable to another.

5.4 From damaged container to provisional identity — Review

MOLECULAR IDENTIFICATION · MOLECULAR IDENTIFICATION LAB · A ROOM · CALLBACK · SEQUENCE

SCENE 42W

Ana Kowalski, who keeps the reference library and standards, signs in a warm sludge jar recovered from the outfall. It has a headspace, an intact seal, and enough material for one destructive run. Okonjo wants a handling order before anybody opens it.

GUIDE 69W

All four of these could be done in the next ten minutes, so nothing here is a clock. What separates them is what is still available afterwards. Ask of each step what it consumes and what it lets escape. The jar came in warm and there is material for one destructive run. Open it in the wrong order and the evidence that answers fastest is the evidence you lose.

ASK 18W

Arrange the four cards from the step that costs the sample least to the step that ends it.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

the volatile fraction of a warm sample sits in the air above it · some measurements leave the sample as they found it and some use it up

VERDICT 86W

A warm sample is already partly in its own headspace. How much is up there depends on how volatile the compound is and on how warm the jar got on the way in. That vapour is the fraction with the shortest life. Break the seal and it is gone in seconds, and no later measurement recovers it. Weighing and logging cost nothing and can be repeated. A spectrum borrows an aliquot and returns it. The destructive run answers best and answers once, so it goes last.

TAKEAWAY 22W

> The volatile fraction has the shortest life of any evidence in the jar, so it is taken while the seal still holds.

END OF THE DAY 14W

Predict where each chemical class will be found and design a mass-balance sampling plan.

Can We Trust the Numbers?

PLAN CARD 49W

On day 8 the first map of the numbers went up, and four systems shut because of it. Some readings ran off the top of the scale, and some were watered down first. Today you say which numbers can carry a decision, and where the next thirty samples go.

BRIEFING 20W

Some concentration results are over-range, some were diluted, and a thin sample map can still miss the places that matter.

WHAT YOU DECIDE 19W

Turn instrument signals into concentrations with calibration, dilution, quality control and a sampling design tied to the reopening decision.

6.1 Undo the dilution

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · BALLPARK

SCENE 46W

Nakamura runs quality assurance. She hands you the raw reading with its preparation log. The instrument reports what was in the vial, not what was in the river. City leaders are comparing your number against a threshold and closing or opening a water system with it.

GUIDE 67W

Five numbers, and two of them belong to different questions: a detection limit and the volume of the bottle. Two others are the volumes before and after the dilution, and only their ratio matters. Ask of each number whether it describes the vial, the river, or the method. A dilution not carried back understates the river tenfold, and nothing about the reported number says it was diluted.

ASK 4W

Estimate the original concentration.

BEHIND THE BUTTON 144W

What the relationship says. The concentration of the solution is the amount dissolved divided by the volume holding it, so amount = C × V. Diluting adds solvent and not solute, which leaves C_original V_sample = C_diluted V_final.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

diluting a sample lowers its concentration by a known factor

VERDICT 74W

Samples are diluted so the signal lands inside the range the method can measure. That factor has to be carried back through before the number says anything about the river. A tenfold dilution reported as measured understates the river ten times over — and the report still looks like a valid result, because nothing about the number says it was diluted. Preparation is part of the answer, not something that happened to it first.

TAKEAWAY 8W

Sample preparation is part of the quantitative result.

6.2 Protect the calibration

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · VERIFY

SCENE 35W

An instrument will return a number for almost anything you put in front of it. Dana Whitfield, the calibration technician, lays out the panel — everything needed to decide whether this reading can be defended.

GUIDE 55W

Lock the absorbance you expect after a one-to-one dilution before you make it. Then make the dilution and read what the instrument returns. The last step costs a verification slot, and spending it is the point. An instrument returns a number for almost anything, and only the comparison says whether this one can be defended.

ASK 15W

What should the absorbance become after dilution, and does the rerun support the dilution model?

TAKES AS READ

Beer–Lambert gives A proportional to concentration when path length and absorptivity are fixed · a 1:1 dilution halves concentration

VERDICT 77W

Beer–Lambert predicts absorbance proportional to concentration while the method remains linear. A 1:1 dilution halves concentration, so an absorbance of 1.34 should fall to about 0.67. The observed 0.68 is inside the validated calibration range and agrees with that prediction. The diluted concentration can then be read from the calibration and doubled for the original sample. This is stronger than extrapolating 1.34 because both the dilution and the response are checked inside the range established by standards.

TAKEAWAY 22W

A measurement outside its calibrated range becomes defensible only after a controlled change moves it into-range and the predicted response is verified.

6.3 Spend thirty laboratory samples where they change the decision

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · CHOICE

SCENE 43W

Reyes runs the water utility. She needs a defensible map of eleven kilometres of river. The field kit screens densely, but it is much noisier than the laboratory. Thirty laboratory analyses must cover the intake decision, movement of the contamination, and quality control.

GUIDE 29W

The field kit already buys coverage. Spend the scarce laboratory runs on the places and checks that can overturn the reopening decision: the intake/endpoints, movement sentinels, and quality control.

ASK 12W

Which design makes the strongest reopening decision with only thirty laboratory samples?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a map is only worth what the decisions it supports are worth

VERDICT 70W

The intake deserves the densest high-quality sampling because reopening turns on it, but putting all thirty samples there leaves transport and background unconstrained. Even spacing all thirty once along the river spreads precision too thin at the decision point and says little about time. A mixed design uses the noisy field kit for coverage, anchors the intake with laboratory analyses, places upstream/downstream sentinels, and spends some runs on quality control.

TAKEAWAY 16W

Decision-focused sampling still needs spatial sentinels and quality control; precision at one point is not representativeness.

6.4 What the bench is running now

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · SPOT

SCENE 33W

Samples queue on the bench all day. Nakamura holds one priority, and changes it as the incident does — after the river result, after the schools ask, after the hold times start expiring.

GUIDE 53W

Run the samples the current priority wants and leave the rest queued. The priority on the bench card is changed during the day and nobody announces it. What is scored is the samples either side of a change, because they are the only ones that show whether the bench is reading the card.

ASK 26W

Run to the priority on the bench card, and keep watching the card. A sample run past its hold time carries an uncertainty nobody can bound.

BEHIND THE BUTTON 91W

Why a laboratory priority changes. Early it is whatever the incident commander is waiting on. Then it is whatever is closest to a hold time, because a sample analysed after its hold time is not a result. Then it is whatever a decision is actually waiting on, which is rarely the same list.

Why the changeover costs. A bench working to the old priority for another twenty minutes spends its one clean instrument on a sample nobody is waiting for, and the sample that was waiting expires quietly in the fridge.

TAKES AS READ

a laboratory works to a stated priority that somebody can change

VERDICT 141W

Three priorities run across the day and each wants a different part of the rack. Anything the commander is waiting on, then anything near its hold time, then anything from a drinking water source. A sample can answer two at once, which is what makes the change cost real. The panel scores the window either side of each change, because most of the rack is wanted by neither priority and is correctly left queued by somebody who has read nothing. The laboratory's own version is the twenty minutes after the schools ask: the priority has changed to drinking water, and the instrument is still working through river samples that were urgent an hour ago. And the cost is invisible afterwards — a sample past its hold time comes out of the fridge looking exactly like a sample that is still valid.

TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

END OF THE DAY 12W

Produce concentration maps whose units, detection limits, and quality controls are explicit.

DAY 7 OF 15

The Dose That Overshot

PLAN CARD 48W

On day 11 the plant dosed the tank from the pH reading, and it shot past the target. This water holds a store of acid that pH alone barely shows. Today you split free acid from total demand, then set a rule that stops the feed in time.

BRIEFING 18W

The starting pH measures only free hydrogen ion, and badly underpredicts how much base the water will consume.

WHAT YOU DECIDE 16W

Use pH, titration and buffer behaviour to choose a neutralisation strategy without overshooting the treatment window.

7.1 How many moles of acid are present?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · BALLPARK

SCENE 36W

The river intake is acidic and the plant is preparing a neutralisation dose. Ferreira measured a one-thousand-litre test volume at pH 4.0 overnight. Novák wants the size of the free hydrogen-ion pool before anybody adds base.

GUIDE 65W

Four numbers, and two of them are the pH reading itself and the concentration in neutral water. A pH is a logarithm, so the reading cannot enter a mole calculation as a concentration. Convert first, then multiply by the volume. Note what the answer covers. This is the free hydrogen ion at this moment. A buffered water can release much more as base goes in.

ASK 11W

How many moles of free H^+ does the pH reading represent?

BEHIND THE BUTTON 121W

What the relationship says. $pH = -\log[H^+]$, so $[H^+] = 10^{(-pH)}$. Then $n = cV$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

pH is a logarithmic scale — one unit is a factor of ten

VERDICT 73W

A pH of 4.0 means $[H^+] = 10^{-4}$ mol/L, not 4 mol/L and not 4 times neutral water. Multiplying that concentration by 1,000 L gives 0.10 mol of free H^+. That number is useful but incomplete. Weak acids and buffers can release more H^+ as base is added, so pH does not measure the full neutralisation demand. A titration does, which is why the later dosing decision cannot come from this calculation alone.

TAKEAWAY 19W

pH is logarithmic, and converting it to moles still captures only the free hydrogen ion present at that moment.

7.2 Read the titration curve

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · DIAGNOSIS

SCENE 37W

The trial dose calculated from the intake pH overshot badly. Ferreira sets her overnight bench titration beside a strong-acid reference curve, with fresh pH 4 and pH 7 calibration checks from the same electrode on the screen.

GUIDE 74W

Six readings, and two of them exist to rule things out: the fresh calibration checks and the strong-acid reference at the same pH. Compare the shape of the trial curve against that reference before choosing. A long plateau and an equivalence point above neutral are both features a strong acid does not have. The trial dose overshot, which means whatever explains the curve also has to explain a demand the starting pH never showed.

ASK 22W

The titration curve is on screen. Which explanation accounts for the plateau, the equivalence point and the dose the trial actually needed?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a weak acid holds most of its acidity undissociated

VERDICT 77W

A titration curve shows how much base the water consumes as pH changes. A long plateau means a buffer is absorbing added base while pH moves slowly. An equivalence point above pH 7 is also diagnostic of a weak acid system. The conjugate base left near equivalence is itself basic. A strong acid reference reaches equivalence sharply and with much less added base. The curve therefore measures total neutralisation demand that the starting pH alone cannot reveal.

TAKEAWAY 10W

Titration is both a quantitative method and a diagnostic fingerprint.

7.3 Control pH without overshoot

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · TRIGGER

SCENE 38W

The base pump is primed and Novák's operators are waiting to open the valve. Intake water is at pH 4.8, her cast-iron line is corroding, and every pump change appears one sampling interval later on the mixed-tank electrode.

GUIDE 86W

One rule, written before the run and fixed the moment you release it. The base feed is pushing the tank up, and anything past pH 7.5 breaches consent from the other side. The catch is delay: a pump command shows on the mixed-tank electrode one reading later, so the tank goes on climbing after the feed is off. Put the line back from 7.5 by that one reading of travel — and not so far back that it fires on water which has shown nothing yet.

ASK 19W

The tank is climbing from 4.8 towards the 7.5 limit. At what pH do you stop the base feed?

TAKES AS READ

the mixed-tank pH reading lags a pump change by one sampling interval · pH is logarithmic, so equal pH steps do not represent equal amounts of hydrogen ion

VERDICT 84W

Neutralisation in a flowing plant has delay. A pump change occurs now, but the mixed sample that proves its effect arrives later. Waiting for the displayed pH to reach the desired final value therefore guarantees overshoot. Staged dosing writes the rule in advance. The feed is slowed while buffer demand can still be measured. It is then stopped early enough that mixing and residual base carry the final reading upward without crossing the limit. The threshold is a control decision, not a retrospective interpretation.

TAKEAWAY 25W

When a process has delay, a control threshold must be set before the final limit because the system keeps moving while the action takes effect.

7.4 Which reactant limits? — Review

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · CALLBACK · BALLPARK

SCENE 41W

The dosing tank takes three moles of C for every two of D, and the tote arrived part-filled. Tomas Brandt, who leads reactions and energy, will not sign a gas-production figure until somebody says which of the two runs out first.

GUIDE 63W

Four numbers, and one of them is the two amounts added together, which no stoichiometry ever uses. Another is the reaction's own ratio rather than a quantity of anything. Ask what each number is a quantity of before you place it. One reactant's amount, put through the ratio, is a claim about the other. The tank is charged from whichever figure gets signed.

ASK 11W

Which reactant limits, and how many moles of D can react?

BEHIND THE BUTTON 135W

What the relationship says. Convert one reactant through the coefficients to the amount of the other it can consume, then take the smaller. Percent yield = actual/theoretical × 100.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a reaction stops when one ingredient runs out

VERDICT 87W

A balanced equation compares particles, so the coefficients are a currency for converting one reactant into the other. Twelve moles of C at three C to two D is a demand for eight moles of D. There are nine on the pad, so C runs out and D is left over. The ceiling is set by the reactant that goes first, and nothing beyond it produces heat or gas. Recovering seven point two moles of reaction against that ceiling is ninety per cent of what was available.

TAKEAWAY 22W

The reactant that runs out first sets the ceiling, and everything past that ceiling is arithmetic about a reaction that cannot happen.

END OF THE DAY 13W

Select a controlled neutralization strategy and determine when buffer capacity will be exhausted.

The Riverbed Is Not Clean

BEFORE THE DAY — HUNT

Six drums on the manifest, four in the yard

The manifest says six drums came off the lorry and the yard holds four. Until the other two are found nobody can say what is in the ground, because a drum nobody has located is a drum that is either in a skip or leaking somewhere nobody is sampling. Find them.

PLAN CARD 46W

On day 14 two weeks of water samples read low, and the sediment cores read high. The metal leaving the water is turning up in the river bed. Today you say which changes could put it back into the water, and what must still be watched.

BRIEFING 12W

Treatment changed where the hazard sits; it did not destroy the metal.

WHAT YOU DECIDE 17W

Show how equilibrium can move the metal between dissolved and solid phases and design monitoring for remobilisation.

8.1 From condition change to new equilibrium

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A ROOM · SEQUENCE

SCENE 42W

After the pH adjustment the dissolved concentration has fallen and the sediment concentration has risen. Camila Ibarra, the sediment analyst, puts both curves on one page — reading only the water is how a reservoir gets declared clean while the contaminant settles.

GUIDE 58W

Each of these four needs something the others supply. Ask what each one cannot be done without. An equilibrium expression cannot be written for chemistry nobody has named. Changing a condition means nothing until the model balances mass. And the last card is the one this stop turns on: a prediction nobody measured against is not a prediction.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

material is conserved — a shift moves it, it does not destroy it

VERDICT 83W

Predicting where a system lands takes both halves of the problem. The relevant reactions have to be identified first, because you cannot write an equilibrium expression for chemistry you have not named. The expressions and the mass balance come next: the expressions say which direction the system moves, and the mass balance says how much of what has to be somewhere. Only then does changing a condition in the model mean anything, and only then can a prediction be compared against a measurement.

TAKEAWAY 13W

> A lower dissolved concentration may mean phase transfer, not removal from the system.

8.2 Interpret the shift

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · PROTOCOL

SCENE 46W

Reservoir conditions are changing as the treatment continues. Operators have removed some solids, a new ligand has entered with an upstream discharge, and pH varies through the day. Stavros, the long-term monitoring officer, points out that next spring nobody will be sampling continuously for remobilised metal.

GUIDE 71W

Four changes to the reservoir, and each pushes the same equilibrium a different way. Ask of each which side it takes something from or adds something to. One of them raises the total dissolved metal while lowering the free metal, which reads as a treatment failure and is not. Next spring nobody will be sampling continuously, so the answer has to be about the environment the sediment will be in then.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a system at equilibrium responds to a change by partly offsetting it

VERDICT 83W

Le Chatelier's principle gives the direction, and each condition pushes a different way. Removing a solid phase from contact takes away the source of further dissolution. A ligand that binds the dissolved metal holds it in solution, so total dissolved concentration can rise even as free metal falls. A pH shift that favours an insoluble hydroxide drives precipitation. A sediment that later meets acidic water can release what it captured. A contaminant locked into a solid this month can therefore return next month.

TAKEAWAY 12W

Treatment plans must anticipate the next environment, not only the current sample.

8.3 Will the holding pond freeze tonight?

SOLUTIONS & WATER CHEMISTRY · WATER INTAKE LABORATORY · A PERSON · BALLPARK

SCENE 51W

The temporary holding pond feeds the treatment pumps. The forecast low is −4 °C; if the pond freezes at the surface intake, untreated water has to be bypassed or the plant shuts. Ferreira measures 1.2 mol of dissolved NaCl per kilogram of water and asks whether chemistry alone keeps it liquid.

GUIDE 71W

Five numbers, and two of them are reference points rather than terms. One is the freezing point of pure water; one is the temperature the pond is at. Another is the number of particles each formula unit contributes, which is where the salt does its work. Ask of each whether the relationship needs it or the comparison does. The answer decides whether this is ordinary chemistry or an unmeasured heat source.

ASK 10W

Should the saline holding pond remain liquid at −4 °C?

BEHIND THE BUTTON 121W

What the relationship says. $\Delta Tf = iKf\,m$, and $T\_freeze \approx 0\ °C - \Delta Tf$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

molality is moles of solute per kilogram of solvent · NaCl contributes about two dissolved particles per formula unit in this estimate

VERDICT 75W

For an ideal solution, $\Delta Tf = iKf\,m$. Sodium chloride contributes roughly 2 dissolved ions, so i is about 2. With Kf = 1.86 °C kg/mol and m = 1.2 mol/kg, the depression is about 4.5 °C. That puts the expected freezing point near −4.5 °C, so liquid water at −4 °C is plausible without any extra heat source. The estimate is imperfect at high salt concentration, but it gives the correct scale and direction.

TAKEAWAY 18W

A colligative-property estimate can decide whether a cold-weather observation is expected chemistry or evidence of another heat source.

END OF THE DAY 16W

Predict how pH and ligands shift dissolved concentration and design monitoring for the metal coming back.

DAY 9 OF 15

The Plume Came Back

PLAN CARD 46W

On day 17 the first vapour is nearly gone, and the city was told the air scare had ended. Then the street monitor climbs again each afternoon. Today you test whether the sun is making a new gas, and move the monitors to where people live.

BRIEFING 18W

The primary volatile has fallen, but an afternoon secondary product is rising downwind after the source was sealed.

WHAT YOU DECIDE 19W

Identify secondary atmospheric products and move monitoring to the places and times that can still change the return-home decision.

9.1 Build a secondary-pollutant pathway

ATMOSPHERE & GASES · AIR & PLUME STATION · A ROOM · SEQUENCE

SCENE 42W

The primary vapour is falling after the source was sealed, but a different compound is rising at the neighbourhood monitor each afternoon. Erik Lindqvist, the dispersion modeller, posts sunlight, oxidant measurements and wind data from the same period beside the chemical traces.

GUIDE 65W

These four are a chain, and each step makes what the next one needs. Ask what has to be present before each card can happen. Radicals cannot attack anything before something creates them. Nothing can be transported before it exists. The primary vapour is falling while a different compound rises each afternoon. The order of this chain tells the monitoring team which compounds to add.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

sunlight carries enough energy to break chemical bonds

VERDICT 79W

Sunlight can start atmospheric chemistry that does not occur at night. Photons break bonds or create reactive radicals, and those radicals rapidly attack other molecules in air. The products can differ in toxicity from what was emitted. A primary vapour can therefore fall while a secondary pollutant rises later and farther downwind. Building the pathway in order links the daytime pattern to a mechanism. It also tells the monitoring team which new compounds and chemical drivers should be measured.

TAKEAWAY 10W

Exposure can peak after the original emission begins to decline.

9.2 Read the day-night pattern

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · DIAGNOSIS

SCENE 46W

The source has been sealed, but the afternoon monitor is climbing again. Yvette Delacroix, on the neighbourhood health desk, must renew the shelter notice today or let it lapse. Two stations have run a full day; their traces do not look like a plume drifting away.

GUIDE 70W

Six readings, and the inert tracer and the shared inlet material are there to rule two candidates out. Compare the whole day at both stations rather than the afternoon at one. A plume that drifted back carries its own tracer with it; an inlet artefact appears at both monitors together. A compound made in the air has a shape: it rises with the sun, lags the maximum, and falls overnight.

ASK 26W

The source is controlled and exposure went up. Which explanation fits the whole day at both monitors well enough to renew or lift the shelter advice?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

a compound blown around by wind has no reason to care what time it is

VERDICT 75W

Photochemistry has a signature. Production needs sunlight, so a compound formed in air rises during the day. The peak can lag the solar maximum because the chemistry takes time. Production then falls overnight as the reactive radicals disappear. Formation in the air also shifts the peak downwind because the parcel reacts while it travels. A compound that is merely transported lacks this coordinated

> day-night and downwind pattern. That makes the shape diagnostic rather than incidental.
>
> **TAKEAWAY** 11W
>
> Atmospheric chemistry must be inferred from coordinated chemical and weather patterns.

9.3 **Monitor the transformed plume**

ATMOSPHERE & GASES · AIR & PLUME STATION · A PERSON · VALUE

SCENE 44W

Residents want to come home, and Varga has the monitoring van idling at the fence. The released compound is down to 3 parts per billion at both monitors. Total oxidant is up fourfold since Tuesday, and afternoon readings remain highest six days after sealing.

GUIDE 48W

Two channels can be added this week, and residents want to come home. Open each measurement and ask whether its result could change that decision. The released compound is already low at both monitors, so more of the same measurement cannot move anything. Buy the channels that can.

ASK 16W

Sunlight is making a secondary product downwind. Which measurements can change the return-home decision this week?

BEHIND THE BUTTON 135W

Why the original compound is no longer the question. It is down to a few parts per billion and the source has been sealed for six days. A measurement that confirms a number everybody already accepts cannot change a decision, however cleanly it is made.

What the oxidant rise is saying. Total oxidant up fourfold, with afternoon readings highest, is the signature of something being formed in the air rather than released from the site. Sunlight drives it, which is why it peaks in the afternoon and why sealing the source did not stop it.

What a return-home decision actually needs. Not the concentration of the thing that was spilled, but of whatever residents will breathe next week. Those are different molecules, and the monitors currently watching for the first are silent about the second.

TAKES AS READ

a monitoring network only detects the compounds it is configured to measure · chemical drivers can explain exposure without being the exposure themselves

VERDICT 84W

The source compound is already low at both existing monitors, so measuring it more precisely does little to test the new hazard. Sunlight and oxidants explain why secondary chemistry should occur, but they are not themselves the exposure. The decisive channel measures the predicted secondary products where people breathe them. A second channel can then measure the chemical drivers to explain and forecast the pattern. This ordering separates evidence that changes the return-home decision from evidence that merely makes the mechanism easier to describe.

TAKEAWAY 21W

> When chemistry transforms a released compound, monitoring must follow the exposure product rather than only the original source or its drivers.

END OF THE DAY 10W

Identify likely secondary products and update monitoring locations and times.

DAY 10 OF 15

Why Is It Still Heating?

PLAN CARD 45W

On day 22 one storage bay is hotter than it was at midnight, weeks after the fire went out. Fire Command wants the cordon down tonight. Today you turn that heating into an energy figure, and say what would prove the heat source is gone.

BRIEFING 21W

The decision is whether the storage bay is slowly cooling from the original fire or still generating heat of its own.

WHAT YOU DECIDE 19W

Distinguish stored heat from continuing reaction and require monitoring that can detect self-heating before release of the storage zone.

10.1 How much energy raises the temperature?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · BALLPARK

SCENE 40W

The visible fire has been out for two hours, but the containment bath beside the damaged storage bay is still warming. Sørensen wants the size of the stored heat load before deciding whether the cooling equipment on site is adequate.

GUIDE 68W

Five numbers, and two of them belong elsewhere: one to a change of phase, one to the surrounding air. Ask of each whether it describes the mass, the material, or the change. Latent heat answers a different question and is much the largest number here. A small warming across a large thermal mass is a great deal of energy. Fire Command is sizing cooling equipment from the answer.

ASK 4W

Estimate the absorbed heat.

BEHIND THE BUTTON 130W

What the relationship says. $q = mc\Delta T$, with q the heat, m the mass, c the specific heat capacity and ΔT the temperature change.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

different materials need different amounts of energy to warm by one degree

VERDICT 77W

The energy needed to change a material's temperature is its mass times its specific heat capacity times the temperature change. A small rise across a large thermal mass can therefore represent a very large amount of energy. Water's high heat capacity makes it a useful containment bath, but it also hides how much energy has entered. Temperature alone says little about the quantity of stored heat. It becomes useful only when connected to mass and heat capacity.

TAKEAWAY 10W

Temperature becomes meaningful when connected to mass and heat capacity.

10.2 Stored heat or continuing reaction?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · DIAGNOSIS

SCENE 40W

Two storage bays were exposed to the same fire. 90 minutes after cooling stopped, Brandt's readings show the east bay cooling while the west bay warms. West-bay off-gas is increasing, and the hottest point remains fixed at the drum stack.

GUIDE 71W

Six readings are on the panel and the east bay is the useful one, because it is the same fire with nothing left running. Compare the two bays reading by reading before choosing. A miscalibration moves both bays together; stored heat can only drain toward cooler surroundings. Rising temperature with rising off-gas at one fixed point is a different claim from either, and Fire Command releases the site on this answer.

ASK 24W

Fire Command needs to know whether this is heat left over from the fire or a reaction still running. Which explanation fits every zone?

BEHIND THE BUTTON 179W

Why the unremarkable readings decide it. The salient reading is what draws attention, and it is usually consistent with several explanations at once, which is why it rarely settles anything. The readings that discriminate are the ones a candidate predicts should have moved and which have not: a normal value is a positive result against every mechanism that would have disturbed it.

How to work the candidates. Take each mechanism and predict the panel it implies before you look at the panel again — which readings it drives, in which direction, and by roughly how much. Then compare. Working that way round is what separates a diagnosis from a rationalisation, because the prediction is made before the data is consulted.

Why only one candidate survives. Several will account for part of the panel, deliberately so, and a partial fit is exactly what a confident wrong answer feels like from the inside. When two remain, look for the reading on which their predictions differ and let it decide. If no reading separates them you have not finished reading the panel.

TAKES AS READ

heat flows from hot to cold and nowhere else on its own

VERDICT 82W

A hot mass can be carrying stored energy from an earlier event, or it can be generating new energy now. After the source is removed, stored heat can only drain toward cooler surroundings. The east bay shows that pattern and provides a control. The west bay instead rises in temperature while off-gas increases at the same fixed location. That combination requires ongoing heat generation. A calibration error would move both bays together, so the diverging trends point to chemistry rather than instrumentation.

TAKEAWAY 11W

Thermal diagnosis requires an energy balance, not a single temperature threshold.

10.3 Control the self-heating risk

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · CHOICE

SCENE 36W

Sørensen wants the site released tonight. Brandt's log shows bay 3 rising from 31 to 44 degrees in 8 hours, the rate of rise itself increasing, and the mass is 4 tonnes with a two-metre depth.

GUIDE 67W

All four options are things somebody could require tonight, and three of them are genuinely useful. Ask of each what it would tell you that you do not already know, and how soon. Self-heating is a race between heat made and heat carried away, so only a trend says which side is winning. One reading at the door cannot show a trend, however good the reading is.

ASK 12W

Fire Command wants the site released tonight. What do you require first?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a trend says something a single reading cannot

VERDICT 83W

Self-heating is a race between the energy a reaction makes and the energy the surroundings can carry away. Only the trend says which side is winning, so a site is safe to release when somebody can watch the trend — not when one reading happens to look fine. Controlling it takes three things that do not stand in for each other: detection early enough to act on, the mechanism so the action is the right one, and enough cooling to shift the balance.

TAKEAWAY 8W

High-consequence thermal systems need detection, understanding, and mitigation.

10.4 pH seven, while the plant keeps feeding

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · HOLD

SCENE 41W

Osei's neutralisation tank has to sit at pH 7 before anything leaves for the river. Acid liquor arrives from the works in batches, his lime dosing pump is the only control, and the analyser on the outfall reads every few seconds.

GUIDE 57W

Hold the tank at pH 7 inside the band on the analyser. The band narrows as the discharge window approaches, because a consent is written on what leaves rather than on what was in the tank an hour ago. The lime pump is your control, and each batch that arrives keeps pushing until the dosing answers it.

ASK 12W

Hold the tank at pH 7 while the works keeps feeding it.

BEHIND THE BUTTON 143W

Why pH is not a quantity you can chase. It is a logarithm, so the same dose of lime moves the tank far more near neutral than it does at pH 3 — the last half unit is where a plant overshoots and finds itself dosing acid back in.

Why the buffer decides how twitchy this is. Water with carbonate in it resists change: the same acid batch moves a buffered tank much less. When the buffer is used up the tank stops resisting and moves fast, which is what an operator means by the tank "going".

Why a batch is not a step in the reading. Liquor arriving is a rate, and it keeps arriving. A tank left alone does not settle at a new pH — it carries on moving until the dosing is set to match what is coming in.

TAKES AS READ

pH measures how acidic or basic a solution is

VERDICT 162W

Each batch here is a step in the rate rather than in the reading. Acid liquor arriving does not drop the tank by a tenth of a unit and stop; it keeps arriving, and the tank keeps falling until the dosing is set to match it. That is why the lime pump is moved to a new rate and held. Two things make pH harder than a temperature. It is logarithmic, so the last half unit near neutral takes a fraction of the lime the first three units took, and an operator who has learned the feel of the tank at pH 4 will overshoot at 6.8. And the buffering changes under you: while there is carbonate in the water the tank resists, and once it is consumed the same batch moves it several times as far. The band narrowing is the consent — what is measured is what leaves the works, not what the tank was doing when the shift started.

TAKEAWAY 12W

A feed you do not answer is not a feed that stops.

END OF THE DAY 13W

Create an energy balance and choose a monitoring plan that detects self-heating early.

DAY 11 OF 15

The Pipe We Damaged

PLAN CARD 48W

On day 26 the city's water main sprang a leak where two different metals are joined. Our own emergency earth cable is clamped across the same length of pipe. Today you find the circuit, work out how much metal is gone, and pick the control that breaks it.

BRIEFING 17W

The intake main fails at a mixed-metal joint that the response itself connected electrically during the emergency.

WHAT YOU DECIDE 20W

Identify the galvanic corrosion circuit, estimate the scale of wall loss, and choose a control that breaks the electrical coupling.

11.1 Find the anode and cathode

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · PROTOCOL

SCENE 38W

At the intake gallery, acidic water is flowing through a section where two different metals are electrically connected. Novák's coating inspection has also found one small breach. She wants the corrosion circuit identified before anyone chooses a repair.

GUIDE 64W

A corrosion cell needs four things at once. Somewhere metal is lost. Somewhere electrons are consumed. A path for the electrons, and a path for the ions. Each situation on the left supplies or concentrates one of those parts. Ask which part before pairing anything. Break any one and the circuit stops. So naming the parts is what decides which repair is a repair.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

oxidation is losing electrons and reduction is gaining them · atomic structure, ions and electron configuration — taken as read

VERDICT 59W

Corrosion is an electrochemical circuit: oxidation at the anode, reduction at the cathode, an electron path through connected metal, and an ionic path through the electrolyte. Two dissimilar metals can increase the galvanic driving force. A coating

> defect concentrates current into a small area, which is why local penetration can be much faster than the average mass-loss rate suggests.
>
> **TAKEAWAY** 9W
>
> Stopping corrosion requires breaking or controlling the full circuit.

11.2 How fast can material disappear?

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · BALLPARK

SCENE 42W

Novák hands over a pipe coupon from the intake gallery — 2.0 kilograms of steel lost per year across 4.0 square metres of exposed surface. The steel density is 7,900 kilograms per cubic metre, and the remaining wall is 6 millimetres thick.

GUIDE 67W

Six numbers, and one of them is the density of water rather than of steel. Another is the wall that is left, which belongs to the comparison and not to the rate. Ask of each whether the relationship needs it or the decision does. Notice what a uniform rate cannot see. The same metal loss concentrated into one coating defect perforates far sooner than the average suggests.

ASK 8W

What uniform wall-loss rate does 2.0 kg/year imply?

BEHIND THE BUTTON 129W

What the relationship says. depth loss rate = mass loss / (density × exposed area). Convert metres per year to millimetres per year.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a rate becomes a decision only when it is compared against a thickness · atomic structure, ions and electron configuration — taken as read

VERDICT 80W

Work out the average wall loss because it sets the scale of the problem. 2 kilograms per year over 4 square metres of steel is about 0.063 mm of depth per year. At that uniform rate, losing a 6 mm wall would take roughly 95 years. That result says uniform corrosion is not the immediate threat. Localised attack is worse because the same total metal loss can be concentrated into a small coating defect and perforate the wall far sooner.

TAKEAWAY 16W

A corrosion rate becomes a decision only once it is a depth compared against a wall.

11.3 Protect the pipeline

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 36W

The thinnest section of intake pipe has only 3 millimetres of wall left. Corrosion has accelerated eightfold, and Novák has cleared the gallery until the first control is chosen. Water is still moving through the line.

GUIDE 62W

Three of the four options act on a real part of the corrosion circuit. One only moves paper. Ask of each which of the four parts it removes, and whether one small defect can defeat it. The water is still moving and three millimetres of wall are left. So the first control has to break the circuit, not merely slow it down.

ASK 18W

The gallery stays cleared until the first control is chosen. Which intervention breaks the corrosion problem most directly?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a circuit stops when any one part of it is broken · atomic structure, ions and electron configuration — taken as read

VERDICT 77W

A corrosion cell needs an anode, a cathode, an electrolyte and an electrical path. The acidic, chloride-rich water makes ion transport easy. The connected dissimilar metals provide an electron path and drive galvanic attack. Electrical isolation breaks that path directly. A coating can help, but one small defect can concentrate attack. Chemistry control also matters, yet it does not remove the metal-to-metal connection. The first intervention should break the circuit while inspection and water chemistry controls follow.

> **TAKEAWAY** 11W
> Corrosion protection is a designed system of materials, environment, and inspection.

END OF THE DAY 14W

Identify the corrosion cell and choose controls that address both chemistry and electrical coupling.

DAY 12 OF 15

Can the Treatment Run Away?

PLAN CARD 46W

On day 31 the pilot vessel sat calm for an hour, then its temperature curve bent upward. The plan is to scale it from 40 litres to 9,000 on Monday. Today you build the feedback loop, and pick a plan whose heat stays below the cooling.

BRIEFING 26W

A treatment chemistry that was quiet in a drum accelerates in the warm pilot vessel, and full scale concentrates both the reacting material and the heat.

WHAT YOU DECIDE 21W

Stress-test the treatment kinetics and define a thermal operating envelope that remains safe when the reaction is more temperature-sensitive than expected.

12.1 What changes the rate?

REACTIONS & ENERGY · REACTION & THERMAL HALL · A ROOM · PROTOCOL

SCENE 40W

The same treatment mixture is quiet in a cool drum and vigorous in the warmed pilot vessel. Moreau lays out concentration logs, temperature records and a catalyst trial on the bench, all taken from the same batch before scale-up begins.

GUIDE 71W

The four changes on the left act at three different places. One is the concentration term. One is the fraction of collisions that clear the barrier. One is the height of the barrier itself. Ask which of the three each change touches before pairing it. One of them changes the thermodynamics and leaves the barrier alone. That is the difference between quiet in a drum and vigorous in a warm vessel.

ASK 10W

Which change affects collision frequency, barrier crossing, or activation energy?

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

kinetics means how fast a reaction proceeds and what controls that rate · a reaction needs molecules to meet with enough energy to react

VERDICT 76W

Rate laws separate concentration effects from the rate constant. If rate = $k[A]^2$, doubling [A] multiplies the rate by four because the concentration term is squared. Temperature works differently: it changes the fraction of molecular encounters able to cross the activation barrier, which changes k. A catalyst also changes the pathway and lowers that barrier. Changing product energy changes thermodynamic driving force, but it does not by itself specify the kinetic barrier or the reaction rate.

TAKEAWAY 15W

A reaction can be favorable yet slow, or fast only after a barrier is lowered.

12.2 Build a runaway feedback loop

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · SEQUENCE

SCENE 37W

The pilot vessel showed a sharp temperature rise during one treatment run. Brandt has posted four events from the log on the control-room board, printed out of order, and wants the causal sequence reconstructed before engineering meets.

GUIDE 70W

These four are one loop, so every card has something before it and something after it. Ask of each what has to be true already for it to happen. Heat has to exist before it can raise a temperature, and a temperature has to rise before a rate constant does. Put the chain in order and a run that looked alarming becomes something a cooling system can be sized against.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a reaction goes faster when it is hotter

VERDICT 75W

It is a feedback loop with an obvious first step. An exothermic reaction releases heat. That heat raises the temperature when removal cannot keep pace. The cooling system removes heat at a rate set by its geometry and the available temperature difference. A higher temperature raises the rate constant. A faster reaction releases the same heat sooner, and the loop closes. Put the chain in order and it stops being alarming and starts being designable.

TAKEAWAY 11W

Runaway risk emerges from coupling reaction rate to imperfect heat removal.

12.3 Define the safe operating envelope

REACTIONS & ENERGY · REACTION & THERMAL HALL · A PERSON · STRESS

SCENE 41W

Osei has the first full-scale treatment batch scheduled before dawn. Bench work stopped at 55 °C, his proposed vessel runs at 60 °C, and the cooling jacket tops out at 40 kilowatts. Heat release above 65 °C has never been measured.

GUIDE 64W

Move the uncertain doubling interval to its pessimistic end before you commit to anything. A plan that survives only at the nominal value is not a plan. The jacket removes 40 kilowatts and no more, bench work stopped at 55 °C, and nothing above 65 °C has ever been measured. Choose the operating plan that still holds at the bad end of the range.

ASK 14W

Which operating plan remains feasible when the reaction accelerates faster than the nominal assumption?

BEHIND THE BUTTON 137W

What the doubling interval means. Reaction rate rises with temperature, and the interval is how many degrees it takes to double. A small interval means the heat release accelerates sharply, so a vessel that is comfortable at 60 °C can be uncontrollable at 68 °C.

Why the jacket is the hard limit. Cooling capacity is fixed at 40 kilowatts. Once the reaction produces heat faster than that, the temperature rises, which makes it produce heat faster still. That is a runaway, and no amount of operator attention reverses it once it starts.

Why the untested region is the whole problem. Above 65 °C there is no measurement at all, so the model there is extrapolation. Committing a full-scale batch to a region nobody has measured is the decision this stop is asking you not to make quietly.

TAKES AS READ

heat generation tracks reaction rate over these compared operating points · a plan is infeasible when heat generation exceeds available cooling

VERDICT 82W

Scale-up cannot be based only on the nominal rate. Heat generation rises steeply with temperature, while heat removal is capped by the jacket and its temperature difference. A plan that barely fits the nominal kinetics can cross the cooling limit if the true temperature sensitivity is stronger. Stressing the uncertain doubling interval exposes that fragility before operation. The conservative plan gives up throughput. It is the only candidate that still has positive cooling margin at the pessimistic end of the stated range.

TAKEAWAY 23W

> A safe operating choice is one that still satisfies the thermal constraint when a key kinetic assumption is wrong in the dangerous direction.

END OF THE DAY 15W

Identify the controlling rate factors and define a safe operating envelope with automatic shutdown triggers.

Gone From the Water Is Not Gone

BEFORE THE DAY — CANVASS

Who actually drank the tap water

The advice went out on Wednesday and the exposure estimate depends on whether people followed it. Ask along the affected streets until enough answers have come back to tell a city that complied from a city that did not, because those two produce very different numbers and the same reassuring press line.

PLAN CARD 47W

On day 38 the treated water reads 98 percent lower, and the city wants to call that a win. The sludge, the waste gas and one new signal have not been counted. Today you close the mass balance, and choose the one pilot question worth four weeks.

BRIEFING 26W

The pilot removes most of the target from the water, but the mass has moved into sludge, off-gas and a poorly characterised pool of transformation products.

WHAT YOU DECIDE 19W

Choose a treatment train from complete mass balance, byproducts, waste fate and operational reliability rather than target removal alone.

13.1 What did the treatment actually do?

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · BALANCE

SCENE 42W

The pilot treatment processed a batch containing 100 kilograms of contaminant. Rafael Delgado, the byproduct analyst, has measured the treated water, sludge and off-gas. City officials are waiting to hear whether the process destroyed the material or moved it into another reservoir.

GUIDE 59W

Read every stream first, because reading costs nothing. Then count only the masses that belong to the same contaminant ledger. The batch went in with 100 kilograms, and treated water, sludge and off-gas have each been measured. Whatever the counted streams do not account for is the missing mass, and that gap is what the destruction claim rests on.

ASK 21W

Close the mass balance. How much material is missing from the measured streams, and does the ledger support a destruction claim?

BEHIND THE BUTTON 146W

What counting a stream claims. Reading is neutral. Adding a stream to the ledger says that the mass in it is the same substance the batch went in with. A breakdown product measured by a method that reports it as the parent compound is the way that claim goes wrong.

Why a gap is not a destruction. Mass that has left the measured streams has gone somewhere: an unmeasured vent, adsorbed on the vessel, or transformed into something none of the three methods sees. Destruction is one explanation for a gap, and the least testable of them.

What the city is actually being told. 'Destroyed' and 'moved into another reservoir' have very different consequences for whoever ends up with the sludge or the air. The ledger is what distinguishes them, which is why the claim has to be made from the numbers rather than about them.

TAKES AS READ

mass is conserved across the treatment process · a concentration decrease in water does not identify where the removed material went

VERDICT 77W

The treated water contains only 6 kg, but that does not mean 94 kg were destroyed. 71 kilograms are measured in sludge and 3 kg appear in off-gas. Those visible streams total 80 kg, leaving a 20 kg term required to close the original 100 kg ledger. The bench already sees an unquantified transformation-product signal, so that missing mass has a plausible destination. The mass balance supports transfer and transformation, not a claim that the contaminant vanished.

TAKEAWAY 19W

> Removal from one stream is not destruction; a treatment claim must conserve mass across all measured products and reservoirs.

13.2 Select a treatment train

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · SEQUENCE

SCENE 33W

Osei spreads four unit processes on the table and wants them in an order before the plant commits. Riverton has one intake and cannot run this experiment twice while the water is off.

GUIDE 66W

These four are not independent machines. Each changes the chemical form that reaches the next. Ask of each card what has to exist before it can do anything at all. Nothing can be settled or filtered until there is a solid. There is no solid until the metal is in the right state at the right pH. Riverton has one intake and cannot run this twice.

ASK 24W

Arrange the four cards from the earliest prerequisite or cause to the latest result. The interface should lock correctly placed cards after each check.

BEHIND THE BUTTON 171W

Why the order is graded whole. A sequence is a claim about dependency: each step is here because the one before it has already happened, or has to have. One transposed pair falsifies that claim wherever it sits, so partial credit would be credit for a sequence that does not work.

Why the ends are the place to start. The first and last cards are constrained by having nothing before or after them, which usually makes them the two you can settle from the scene alone. Pinning them collapses the remaining arrangements sharply, and what is left in the middle is where the actual judgement lives.

What the rail can be ordered on. Time is the usual axis, but it is not the only one. An order can run on cost, on risk, or on reversibility — every observation that leaves the sample as it was, before any that consumes it. On those boards a chronological reading finds nothing, because none of the steps has to happen at a particular hour.

TAKES AS READ

a filter can only remove something that is already a solid

VERDICT 62W

Oxidation must first create the oxidation state that forms the intended insoluble hydroxide. The pH adjustment then drives precipitation. Settling and filtration remove the resulting solids. A selective ion-exchange or contaminant-specific adsorptive medium belongs last to polish the dissolved fraction that remains. The choice of polishing media must match the ion and water chemistry; generic carbon is not a universal dissolved-metal treatment.

TAKEAWAY 16W

A treatment train is ordered by the chemical form each stage creates for the next one.

13.3 Choose the safest pilot program

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 42W

The pilot has room for one question, four weeks, and a tenth of the full-scale flow. Nakamura marks three of the four proposals as resting on the same unmeasured quantity, and the full plant is committed in June whatever the pilot says.

GUIDE 69W

Every method on the table removes the target compound, which is why all of them are on the table. So a pilot measuring removal cannot see the difference it was run to find. Ask of each option what it would tell you that the others would not, and whether it would change the June decision. Judged on removal alone, the winner is whichever method makes the most byproduct fastest.

ASK 12W

All three methods claim removal. What does the pilot have to measure?

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

a method that removes a compound has to put it somewhere

VERDICT 88W

Every method on the table removes the target — that is why they are on the table. What separates them is what they leave behind: transformation products, residuals, and the lifecycle of the waste. A pilot that measures only the target compound cannot see the difference it was run to find, and judged on removal alone it will pick whichever method makes the most byproduct fastest. The right question is the one that would change the decision, and here that means analysing for the products each mechanism predicts.

TAKEAWAY 11W

> The best treatment minimizes total hazard, not merely one measured concentration.

END OF THE DAY 14W

Choose a treatment train using contaminant removal, byproduct formation, waste fate, and operational reliability.

Are We Ready to Reopen?

PLAN CARD 43W

On day 45 the treated water sits just under the release limit, and the first check on the key sample failed. One peak still has two possible names. Today you close those gaps, then decide on release, release with rules, or a hold.

BRIEFING 20W

The last identity question, sampling representativeness and measurement uncertainty all have to be resolved before the distribution network is pressurised.

WHAT YOU DECIDE 23W

Close the last analytical and sampling gaps and make a release decision using a predefined decision band rather than the best estimate alone.

14.1 Name the last unresolved treatment product

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · CHOICE

SCENE 40W

The last unresolved treatment-product peak has two candidate formulas that differ by one oxygen atom. Chromatography cannot separate the identities, but Okonjo has the molecular-ion measurement from that fraction. The release panel must know which product the water actually contains.

GUIDE 22W

The candidates differ by one oxygen atom. Choose the measurement that responds directly to molecular mass rather than retention time or abundance.

ASK 21W

Two candidate transformation products differ by one oxygen atom. Decide which mass-spectral feature can distinguish them before release paperwork is signed.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

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TAKES AS READ

a molecule's formula fixes its mass

VERDICT 80W

A mass spectrometer sorts ions by mass divided by charge. For a singly charged molecular ion, the molecular peak therefore reports molecular mass directly. 1 extra oxygen changes that mass by about 16 units, far larger than the ambiguity in the chromatographic retention time. Peak area instead measures response and abundance, not identity. The mass result supplies a different physical constraint from the separation. That is why it can break a tie between two candidates that chromatography alone cannot distinguish.

TAKEAWAY 20W

> A release limit applies to a chemical identity, so resolving one formula can be as important as measuring its concentration.

14.2 Close the verification gaps

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A PERSON · PROTOCOL

SCENE 39W

A number can be perfectly valid and still fail to answer the question that was asked. Nakamura carries the evidence package in herself — the distribution network is about to be brought back up to pressure on its strength.

GUIDE 65W

The left column is four separate weak points in a verification package. Each has its own remedy. Pair them by asking what specifically was not tested. Sampling only the plant leaves the endpoints untested. Sampling after long flushing measures water nobody receives. A detection limit above the release limit cannot test the standard at all. The network goes back up to pressure on this evidence.

ASK 17W

Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

a sample tells you about the place and the moment it was taken

VERDICT 80W

Verification has several separate weak points and each has its own remedy. Sampling only the treatment plant leaves the endpoints and vulnerable zones untested, and those are where people actually drink. Sampling after long flushing measures water nobody will receive, so the flushing has to be documented and realistic use conditions sampled. A detection limit above the release limit cannot test the standard at all. And one laboratory doing both the optimisation and the verification has marked its own work.

TAKEAWAY 15W

A compliant number is meaningful only if the sampling and method address the actual decision.

14.3 Approve, condition, or hold

QUANTITATIVE ANALYSIS · QUANTITATIVE ANALYSIS & QA · A ROOM · CLOUD

SCENE 46W

Distribution pumps are idle while the board meets. The best estimate is 9 against a limit of 10, and the prewritten decision band extends two units either side. Reyes has operators waiting at the valves, and the city has already been given two different reopening times.

GUIDE 73W

The band on screen is every result the measurement permits, not the one most likely — nine against a limit of ten, give or take two, so part of it is over the limit. Two pieces of follow-up work are available, and each does something different to the band: watch what happens to its middle and to its width separately. Apply what you would authorise, then say whether the release criterion is met.

ASK 16W

Can the entire predefined decision band be brought below the limit, or only its best estimate?

BEHIND THE BUTTON 148W

What the band is. It is the guard band the team chose in advance for this decision, based on the method and sampling uncertainty. It is not a statement that every value inside is equally likely or that values outside are impossible. Its purpose is operational: if the upper edge crosses the limit, the unconditional release rule has not been met.

Moving the centre is not the same as narrowing uncertainty. A rerun that identifies a bias can move the estimate; more representative sampling or a better method can reduce uncertainty. A boundary decision needs both effects tracked separately.

What is riding on the difference. Distribution pumps are idle and the city has already been given two reopening times. A decision taken on the middle alone reopens the network with a real chance of being over the limit, and the second retraction costs more than the first did.

TAKES AS READ

the displayed width is the uncertainty band used for this decision · a central value below a limit does not pass when the supported range crosses the limit

VERDICT 80W

The initial decision band is 9 ± 2, so its upper edge reaches 11 and fails the prewritten rule. An independent rerun can correct the centre without proving the original spread was too wide. Representative endpoint sampling can narrow the band without necessarily moving the centre. In this authored case neither action alone clears the boundary; together they yield 8.6 ± 1.0, whose upper decision edge is 9.6. That satisfies the rule that was chosen before the answer was known.

TAKEAWAY 23W

> Changing a best estimate and reducing uncertainty are different operations; a boundary decision must use the uncertainty rule written before the result arrives.

END OF THE DAY 14W

Make a transparent release decision with conditional monitoring and explicit treatment of borderline results.

DAY 15 OF 15

Sign the Water Back On

PLAN CARD 49W

Seven weeks in, the state reviewer is at the table and the mayor wants one word tonight. The water meets the rule, but the river bed is still dirty and one product is still a guess. Today you sort those claims, audit one last number, and sign or refuse.

BRIEFING 21W

The final board must separate what is established, what remains provisional, and what is safe only while a control keeps working.

WHAT YOU DECIDE 18W

Deliver the claim-by-claim evidence package and decide whether Riverton can reopen under explicit controls, triggers and long-term monitoring.

15.1 Disposition the final chemical claims

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A ROOM · PROTOCOL

SCENE 45W

The city wants one sentence tonight and Mbeki, the public briefing officer, has to put it on the screen. The chemistry does not support one — collapsing everything into a single green light is how a site gets reoccupied above a problem nobody wrote down.

GUIDE 58W

At the end of a release the evidence sits in four different states at once. Each state calls for a different sentence. Ask of each claim what would have to happen for it to become settled. Another method? A validated standard? Continued funding? A chemistry trigger? Collapsing four states into one is how the sentence stops being true.

ASK 31W

Every claim here carries its own uncertainty, and no two carry the same amount. Match each situation to the most scientifically justified interpretation, control, or response. Each choice is used once.

BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

TAKES AS READ

evidence comes in strengths, and a claim inherits the weakest one under it

VERDICT 86W

At the end of a release the evidence is in several states at once, and each calls for a different recommendation. An identity confirmed by independent methods with controls is established and can be closed. A transformation product that is plausible but has no validated standard is provisional, and saying so is what keeps the record honest. Filtered water meeting criteria over contaminated sediment is a control that has to keep being funded. And corrosion held inside a narrow chemistry window is conditional operation with triggers.

TAKEAWAY 9W

Chemical readiness is claim-by-claim, not a single green light.

15.2 Audit the electrochemical removal from the charge passed

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · BALLPARK

SCENE 35W

The electrochemical polishing cell has run for 6 hours at 40 amperes. Before Osei trusts the meter on the water leaving it, he wants to know how much metal that run should have taken out.

GUIDE 69W

Five numbers, and two of them are the same duration in hours and in seconds. Another is the electrons each metal ion needs, which is a property of the metal. Check the units the constant is written in before choosing anything. What comes out is an upper bound rather than a removal. Side reactions carry some of the current. Measured removal below it is efficiency; above it is impossible.

ASK 15W

What upper bound does 6 hours of electrolysis at 40 A place on metal removal?

BEHIND THE BUTTON 119W

What the relationship says. $Q = It$ and $n_{metal} = Q/(zF) = It/(zF)$.

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

current is charge per unit time · a divalent metal ion needs two electrons to become neutral metal

VERDICT 73W

Current times time is charge. Dividing charge by the Faraday constant gives moles of electrons, and a divalent metal needs two electrons per metal ion. 6 hours at 40 A passes 864,000 C, or about 8.95 mol of electrons. The theoretical maximum is therefore about 4.48 mol of metal. Real removal is lower when side reactions carry some current. Comparing measured removal with this bound gives current efficiency and catches impossible treatment claims.

TAKEAWAY 13W

Faraday’s law turns measured charge into a quantitative upper bound on electrochemical removal.

15.3 Sign the reopening order

TREATMENT & INFRASTRUCTURE · PILOT TREATMENT PLANT · A PERSON · CHOICE

SCENE 48W

The state reviewer has accepted the package. Water at the representative endpoints meets the release rule. The sediment is still under management, one transformation product is still provisional, and the intake main is safe only inside a chemistry window. The mayor puts the order in front of you.

GUIDE 34W

Separate the drinking-water decision from the unresolved reservoirs. The water pathway now passes its rule. The remaining question is whether the controls and monitoring that made that true stay attached to the reopening order.

ASK 15W

Use the final evidence package to choose the action that the science can actually support.

BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

TAKES AS READ

representative endpoint water satisfies the prewritten release rule · the treatment and corrosion controls are operating inside their validated ranges

VERDICT 96W

The treated water now passes the prewritten release rule at the representative endpoints. An open-ended hold is therefore no longer supported by the evidence. But three hazards are still open. The sediment stays contaminated, one transformation product has no validated standard, and corrosion returns to the intake main if the chemistry drifts out of range. Reopening with no conditions would erase those limits from the record that documents them. A staged reopening keeps

> them attached. Monitoring of water, sediment and air continues, and a failed chemistry or corrosion trigger puts the hold back on by itself.
>
> **TAKEAWAY** 16W
>
> A defensible reopening is conditional on the controls and monitoring that make the exposure pathway safe.

END OF THE DAY 11W

Deliver a claim-by-claim chemical evidence package and a long-term monitoring plan.

THE CLOSING CARD

The first Riverton valve opened on day sixty-one. The network came back in stages: the treatment conditions stayed inside their validated window, endpoint samples stayed below the release rule, and the hospital tanker was the first one stood down. By evening, taps were running across the city.

The order was not an all-clear. The riverbed still holds contamination, the secondary product remains on the watch list, and the intake main keeps automatic chemistry and corrosion triggers. Four monitoring stations stayed funded after the emergency trailers left, because those open questions were written into the reopening instead of being edited out of it.

You did not make the contamination disappear. You separated the mixed release into things that could be measured, caught a laboratory result that was wrong in the same way twice, followed material from water into sediment and through treatment, and refused to turn a passing number into a permanent promise. Riverton is drinking its own water again because the conditions behind that decision are still visible.

Card text only — no options, boards or answers. 11,253 words of card prose across 52 stops. Generated from `themes/contamcity` by `tools/make-cardbook.mjs`.